AP(19.2)(33) 10.16.2026

AV node → atrioventricular node → is a 2nd clump of specialized cells → located in the inferior portion of the right atrium within the ~~an~~ atrioventricular septum!

→ The ~~se~~ septum prevents the impulse from spreading → directly to the ventricles without passing through the AV node

There is a dramatic pause/critical pause before the AV node depolarizes & transmits the impulse to the atrioventricular bundle

This delay in transmission is partially ~~attributed~~ attributable to the small diameter of the ~~cells~~ cells of the node → which ~~allows~~ slow the impulse.
Also conduction between nodal cells is less efficient than between contracting cells.

oh why?

→ These factors - mean that it takes the impulse approx 100 ms to pass through the node.
The pause is critical ~~for~~ it allows the atrial cardiomyocytes complete their contraction that pumps blood into the ventricles before the impulse is transmitted to the cells of the ventricle itself.

AV node can transmit impulses max @ ~~200~~ 220 per minute. This establishes → the typical max heart rate in a healthy young individual.
High rate, may ~~stop~~ be ~~en~~ ineffective at pumping blood.

(Bundle of His)

Atrioventricular ~~Bundle~~ Bundle
Bundle branches & Purkinje Fibers

Arising from the AV node → { the atrioventricular bundle }